Yasir Arafat

Mechanical Engineering Student
HAN University of Applied Sciences

Langswater 76, 1069TR, Amsterdam, Netherlands

(+31) 687 738 410
arafyasir2023@gmail.com
https://yasirarafat.vercel.app
linkedin.com/in/md-yasir-arafat-bd

EDUCATION

BSc Mechanical Engineering
HAN University of Applied Sciences
Arnhem, Netherlands
2024 – Present

CERTIFICATIONS

- Safety for Operational Supervisors VCA

CAD & SIMULATION

- SolidWorks CAD
- FEM Simulation
- Stress, strain & deformation
- Buckling & factor of safety
- Assembly & weldment design
- Technical drawings / GD&T

PROTOTYPING & MFG

- FDM 3D Printing (PLA)
- Cura slicer & support analysis
- Laser cutting
- Lathe & milling machining
- Mechanical assembly

OTHER

- Prototype testing & validation
- Technical reporting
- Team collaboration

LANGUAGES

English C1
Bangla Native

PROFILE

Mechanical Engineering student with hands-on experience in SolidWorks CAD, FEM simulation, and prototype manufacturing. Across three industry-connected projects I have worked with 3D printing, laser cutting, lathe machining, and iterative design for manufacturability. Comfortable taking ownership of tasks, working independently, and contributing to multidisciplinary teams.

EXPERIENCE

Project Engineer *Sep 2025 – Jan 2026*

Fydro Baking Stones

– Worked as a Project Engineer in a multidisciplinary team at Fydro, collaborating with fellow engineering students to develop a solution for managing Fydro's annual 500 tons of GFRC (Glass Fiber Reinforced Concrete) production residue. Developed solutions to repurpose the residue into a new product, contributing to a closed-loop, sustainable production process.

PROJECTS

CORDS VTS: Cyclist Safety Dummy System *Feb 2026 – Jun 2026*

– Designed articulated dummy components in SolidWorks with a focus on plastic component design and functional assembly.
– Optimised every part for 3D printability: adjusted print orientation, minimised overhangs and support material through Cura slicer analysis.
– Manufactured prototype parts using FDM 3D printing, laser-cut EVA foam, and PVC structural components with magnetic attachment systems.
– Iterated through rapid design-test-refine cycles to meet functional and reusability requirements.

Mechanical Battery: Planetary Gear Energy Storage *Feb 2025 – Jun 2025*

– Designed a planetary gear-based mechanical energy storage system in SolidWorks, including rotational assemblies and structural component layout.
– Manufactured components using lathe machining, FDM 3D printing (PLA springs), and laser-cut wooden housing.
– Tested, analysed, and documented system efficiency, achieving ~33% measured efficiency against a theoretical ~55%.

Case Project: Redesign of Industrial Vibrating Conveyor Structure

Sep 2025 – Jun 2026

– Redesigned an industrial vibrating conveyor structure using SolidWorks CAD and FEM analysis to improve structural stability and reduce deflection.
– Performed static, frequency, and buckling simulations to evaluate stress distribution, vibration behaviour, and structural safety under operational loads.
– Validated simulation results and optimized the frame design for improved performance and durability.